

Individual FoundationPose Setup & Debugging Guide

DISCLAIMER: It is assumed the user has already installed docker and set it with admin permissions

Step 0: Setup docker to run without sudo

Unset

```
sudo groupadd docker
sudo usermod -aG docker $USER
newgrp docker
docker ps
```

1. Initial Setup

Step 1: Clone The Repository (if you haven't done so already)

Unset

```
https://github.com/JustinLungu/FoundationPose-BachelorThesis.git
cd FoundationPose-BachelorThesis
```

Step 2: Download Necessary Weights

- Go to [Google Drive](#)
- Download the two folders, extract,, and place them under the `weights/` folder.
- Ensure the folder names match **2024-01-11-20-02-45** (scorer) and **2023-10-28-18-33-37** (refiner).
- Ensure folder structure is: `weights/2024-01-11-20-02-45/config.yml` (adjust if necessary).

Step 3: Download Demo Data

- Go to [Google Drive](#)

- Download, extract, and place the data under `demo_data/` folder.

Step 4: Modify `run_container.sh`

- Edit `docker/run_container.sh` and update the `PROJECT_DIR` variable to match your specific directory.

2. Install & Configure Docker with NVIDIA Support

Step 5: Install NVIDIA Container Toolkit

Follow the steps at [NVIDIA Container Toolkit Guide](#) or use the commands below:

```
Unset
# Add NVIDIA repository
curl -fsSL https://nvidia.github.io/libnvidia-container/gpgkey | sudo gpg
--dearmor -o /usr/share/keyrings/nvidia-container-toolkit-keyring.gpg \
  && curl -s -L
https://nvidia.github.io/libnvidia-container/stable/deb/nvidia-container-to
olkit.list | \
  sed 's#deb https://#deb
[signed-by=/usr/share/keyrings/nvidia-container-toolkit-keyring.gpg]
https://#g' | \
  sudo tee /etc/apt/sources.list.d/nvidia-container-toolkit.list

# Install NVIDIA Container Toolkit
sudo apt update
sudo apt install -y nvidia-container-toolkit

# Restart Docker
sudo systemctl restart docker
```

Step 6: Configure Docker to Use NVIDIA Runtime

```
Unset
sudo nano /etc/docker/daemon.json
```

Ensure the file contains:

```
Unset
{
  "runtimes": {
    "nvidia": {
      "path": "nvidia-container-runtime",
      "args": []
    }
  },
  "default-runtime": "nvidia"
}
```

Save the file (ctrl+O, enter, ctrl+X) and restart Docker:

```
Unset
sudo systemctl restart docker
```

Verify the configuration:

```
Unset
docker info | grep -i runtime
```

Expected output:

```
Unset
Runtimes: io.containerd.runc.v2 nvidia runc
Default Runtime: nvidia
```

Step 7: Verify NVIDIA Support Inside Docker

```
Unset
docker pull nvidia/cuda:12.4.1-base-ubuntu22.04
```

```
docker run --gpus all --rm -it
nvidia/cuda:12.4.1-base-ubuntu22.04 bash

nvidia-smi

exit
```

Expected output: **GPU details** inside the container.

3. Install & Run FoundationPose

Step 8: Build the FoundationPose Container

```
Unset
cd FoundationPose-BachelorThesis
docker build -t foundationpose-with-docker:latest -f
FoundationPose/docker/Dockerfile FoundationPose
```

Step 9: Run the FoundationPose Container

```
Unset
bash docker/run_container.sh
```

If the container **exits immediately**, check logs:

```
Unset
docker logs foundationpose
```

Step 10: Setup FoundationPose Container from inside

If it's the first time you launch the container, you need to build extensions. Run this command *inside* the Docker container.

Unset

```
bash build_all.sh # use this only the first time you enter the container
```

FoundationPose is Now Running Successfully!

5. Run model-based demo

Step 11: Checking that the demo pipeline is running

Inside the FoundationPose container run:

Unset

```
python run_demo.py  
exit
```

6. Run model-based linemod demo

Step 11: Setup the virtual environment

First make sure you have all the required dependencies to run all pipelines in the project, including the population of the repository:

Unset

```
cd ..  
conda env create -f environment.yml  
#OR  
pip install -r requirements.txt
```

Step 12: Populate the project completely

Unset

```
python populate_pipelines.py
```

To download all the necessary data, including the Linemod dataset, such that all pipelines can be run successfully.

Step 13: Populate the project completely

Now we can go back in the container to test that linemod pipeline works correctly:

Unset

```
cd FoundationPose/docker  
bash run_container.sh  
python run_linemod.py
```